

Parameter Configuration.

```
Dropout = 0.1
SEED = 42
layers = 2
optimizer='adam'
loss='mse'
WINDOW=100
Units=32
Epochs=20
Patience=10
batch_size = 32
5 Year Data (15m)
RMSE (Root Mean Squared Error)
MAE (Mean Absolute Error)
```

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 100, 32)	4,352
dropout (Dropout)	(None, 100, 32)	0
lstm_1 (LSTM)	(None, 32)	8,320
dropout_1 (Dropout)	(None, 32)	0
dense (Dense)	(None, 1)	33

Total params: 12,705 (49.63 KB)
Trainable params: 12,705 (49.63 KB)
Non-trainable params: 0 (0.00 B)

1 Round

```
Epoch 1/20
2325/2325 ————— 172s 72ms/step - loss: 0.0052 -
val_loss: 1.1193e-04
Epoch 2/20
2325/2325 ————— 207s 75ms/step - loss: 8.1821e-04 -
val_loss: 1.0943e-04
Epoch 3/20
2325/2325 ————— 197s 72ms/step - loss: 3.9236e-04 -
val_loss: 7.9676e-05
Epoch 4/20
2325/2325 ————— 166s 72ms/step - loss: 3.1139e-04 -
val_loss: 6.2014e-05
Epoch 5/20
2325/2325 ————— 202s 72ms/step - loss: 3.0369e-04 -
val_loss: 1.0057e-04
Epoch 6/20
2325/2325 ————— 201s 71ms/step - loss: 2.9663e-04 -
val_loss: 6.6845e-05
Epoch 7/20
2325/2325 ————— 172s 74ms/step - loss: 2.8702e-04 -
val_loss: 1.0205e-04
Epoch 8/20
2325/2325 ————— 202s 74ms/step - loss: 2.8801e-04 -
val_loss: 5.5246e-05
Epoch 9/20
2325/2325 ————— 202s 74ms/step - loss: 2.8713e-04 -
val_loss: 8.5794e-05
Epoch 10/20
2325/2325 ————— 202s 74ms/step - loss: 2.8526e-04 -
val_loss: 4.7350e-05
Epoch 11/20
2325/2325 ————— 201s 74ms/step - loss: 2.8067e-04 -
val_loss: 5.0043e-05
Epoch 12/20
2325/2325 ————— 195s 71ms/step - loss: 2.8425e-04 -
val_loss: 5.7618e-05
Epoch 13/20
```

2325/2325 ————— 201s 70ms/step - loss: 2.8391e-04 -
val_loss: 5.8393e-05

Epoch 14/20

2325/2325 ————— 209s 73ms/step - loss: 2.8060e-04 -
val_loss: 4.9842e-05

Epoch 15/20

2325/2325 ————— 195s 71ms/step - loss: 2.7925e-04 -
val_loss: 4.6384e-05

Epoch 16/20

2325/2325 ————— 204s 71ms/step - loss: 2.7522e-04 -
val_loss: 4.6446e-05

Epoch 17/20

2325/2325 ————— 173s 74ms/step - loss: 2.7556e-04 -
val_loss: 5.3289e-05

Epoch 18/20

2325/2325 ————— 202s 74ms/step - loss: 2.7513e-04 -
val_loss: 5.0146e-05

Epoch 19/20

2325/2325 ————— 202s 74ms/step - loss: 2.7520e-04 -
val_loss: 4.3062e-05

Epoch 20/20

2325/2325 ————— 202s 74ms/step - loss: 2.7072e-04 -
val_loss: 5.6854e-05

Restoring model weights from the end of the best epoch: 19.

2325/2325 ————— 44s 19ms/step - loss: 1.1904e-05

MSE Train: 1.2833193977712654e-05

773/773 ————— 14s 19ms/step - loss: 2.5493e-05

MSE Test: 4.306167102186009e-05

773/773 ————— 15s 19ms/step - loss: 8.6435e-05

MSE Val: 7.923972589196637e-05

Test Set RMSE (Original Scale): 0.000906 EURUSD

Test Set MAE (Original Scale): 0.000593 EURUSD

Validation Set RMSE (Original Scale): 0.001229 EURUSD

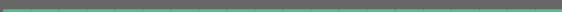
Validation Set MAE (Original Scale): 0.000829 EURUSD

Training Set RMSE (Original Scale): 0.000495 EURUSD

Training Set MAE (Original Scale): 0.000356 EURUSD

4 Round (with GPU)

Epoch 1/20

2325/2325  **30s** 11ms/step - loss:

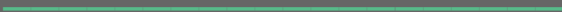
0.0080 - val_loss: 1.2560e-04

Epoch 2/20

2325/2325  **28s** 12ms/step - loss:

9.2954e-04 - val_loss: 1.1865e-04

Epoch 3/20

2325/2325  **27s** 12ms/step - loss:

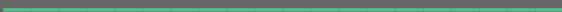
4.3318e-04 - val_loss: 8.7209e-05

Epoch 4/20

2325/2325  **39s** 11ms/step - loss:

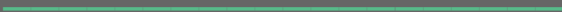
3.1551e-04 - val_loss: 7.7696e-05

Epoch 5/20

2325/2325  **26s** 11ms/step - loss:

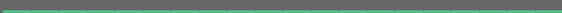
3.0316e-04 - val_loss: 5.8032e-05

Epoch 6/20

2325/2325  **43s** 12ms/step - loss:

2.9941e-04 - val_loss: 1.3675e-04

Epoch 7/20

2325/2325  **41s** 12ms/step - loss:

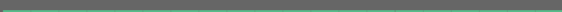
2.9538e-04 - val_loss: 8.4674e-05

Epoch 8/20

2325/2325  **39s** 11ms/step - loss:

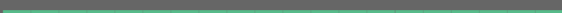
2.8922e-04 - val_loss: 9.2040e-05

Epoch 9/20

2325/2325  **26s** 11ms/step - loss:

2.8367e-04 - val_loss: 5.6847e-05

Epoch 10/20

2325/2325  **43s** 12ms/step - loss:

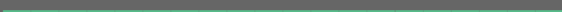
2.8922e-04 - val_loss: 4.6677e-05

Epoch 11/20

2325/2325  **39s** 11ms/step - loss:

2.7941e-04 - val_loss: 8.7909e-05

Epoch 12/20

2325/2325  **41s** 11ms/step - loss:

2.8404e-04 - val_loss: 9.1578e-05

Epoch 13/20

2325/2325  **26s** 11ms/step - loss:

2.8344e-04 - val_loss: 4.7071e-05

Epoch 14/20

2325/2325  **41s** 11ms/step - loss:

2.8453e-04 - val_loss: 6.2632e-05

Epoch 15/20

2325/2325  **41s** 11ms/step - loss:

2.8079e-04 - val_loss: 5.8865e-05

Epoch 16/20

2325/2325  **41s** 11ms/step - loss:

2.7615e-04 - val_loss: 5.6188e-05

